STORAGE



Increasing Block Sizing Changes Data Placement Economics



Mike James (credit to Liam Parker)

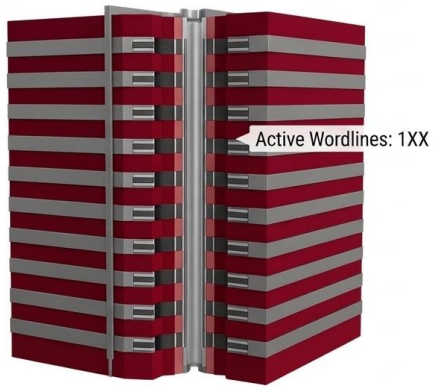
SANDISK

NAND Erase Block Growth is >250%!

While ever-increasing NAND block sizes accommodate higher capacities, tradeoffs can significantly change the cost structure for implementing small vs large unit data placement. These tradeoffs have complex performance and cost implications due to changes in WAF, SLC cache required, and data protection schemes.

Generation X – 1XX Layer TLC

3 pages per String Layer
X
1XX Layers
X
4 Planes
X
16KB Page Size
=
~22-36MB Per Erase Block





Generation Z – 3ZZ Layer QLC

4 pages per String Layer
X
3ZZ Layers
X
4 Planes
X
16KB Page Size
=
~100-140MB Per Erase Block

NAND Erase Block Growth Over 2 Generations and 1 bit per cell is 250%+!

Use Data Placement to Manage WAF

Data placement needed to manage WAF with minimized extra blocks

Direct write* with minimal extra blocks a key technology enabler.

- Emerging high-capacity SSDs and NAND die populating those SSDs leverage legacy HDD workloads to get to improved cost structure.

SLC cache size grows as NAND block size grows.

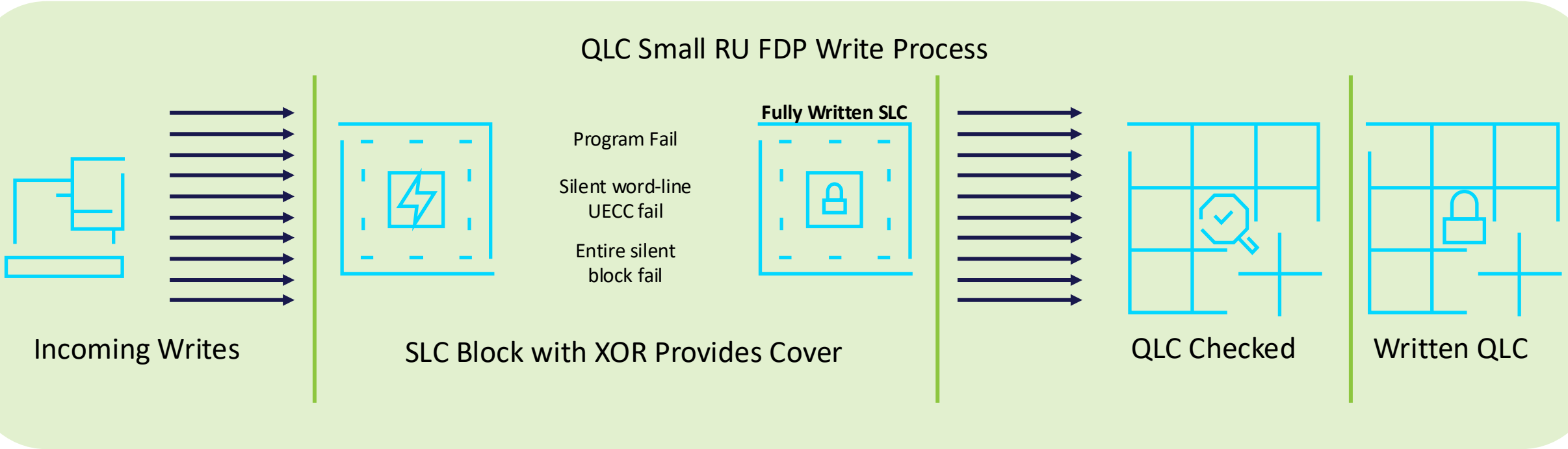
- Small-RU FDP requires an SLC Cache to achieve write reliability.
- Scaling tenants as capacity grows means scaling open blocks

Large-RU data placement and conventional becomes cheaper than small-RU FDP

- As block sizes grow, the relative cost per GB of small-RU FDP compared to large-RU data placement or conventional SSD inverts.

***Direct Write** is our name for a new generation of QLC programming method where the first pass QLC program is readable, which completely removes the need for caching the host write data in SLC

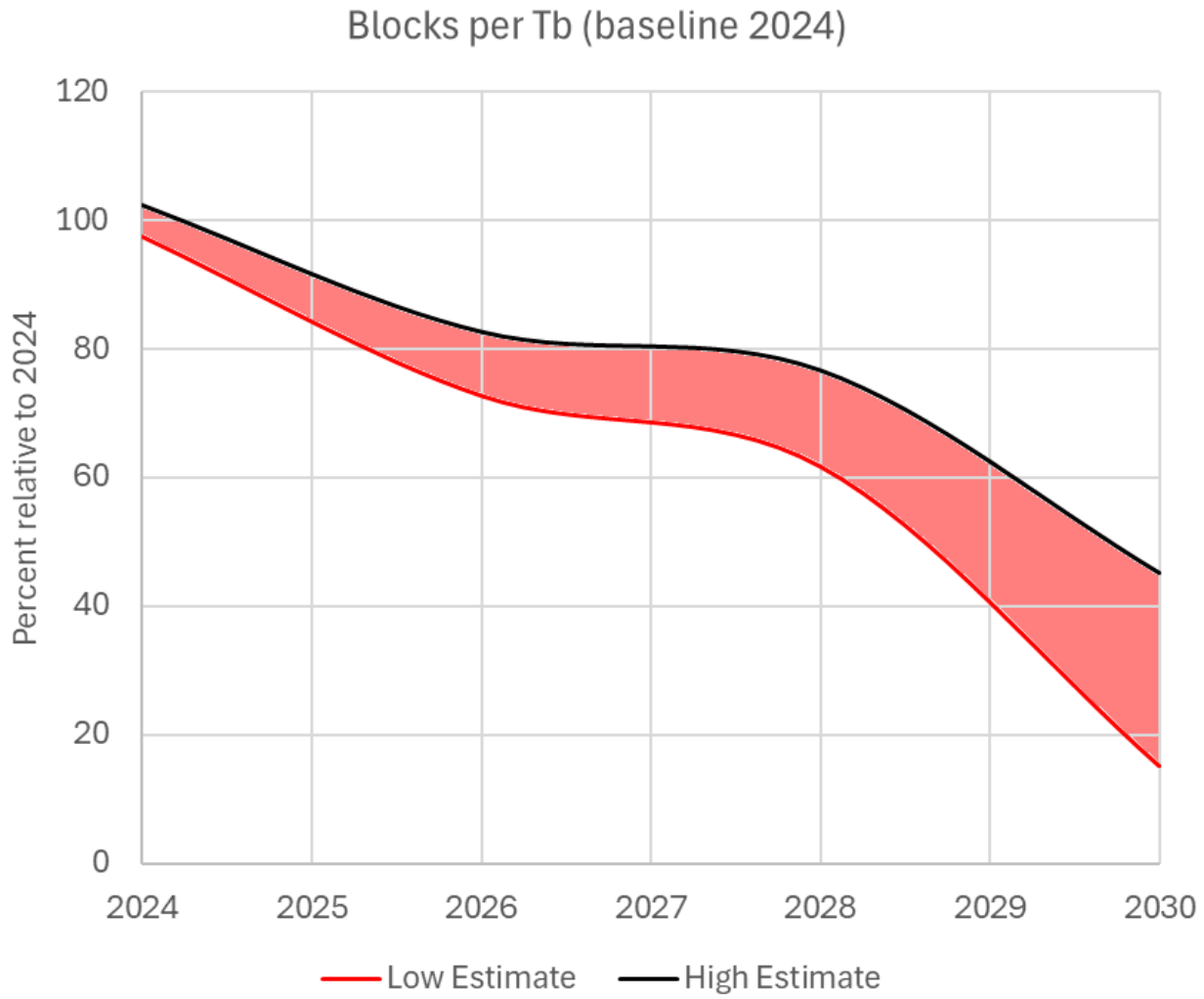
SLC Cache Sizing with Temporary XOR : QLC Small RU FDP



The SLC pool also requires over-provisioning due to “random write” effect of different open RUHs being written at different rates

Ever Increasing NAND Block Sizes

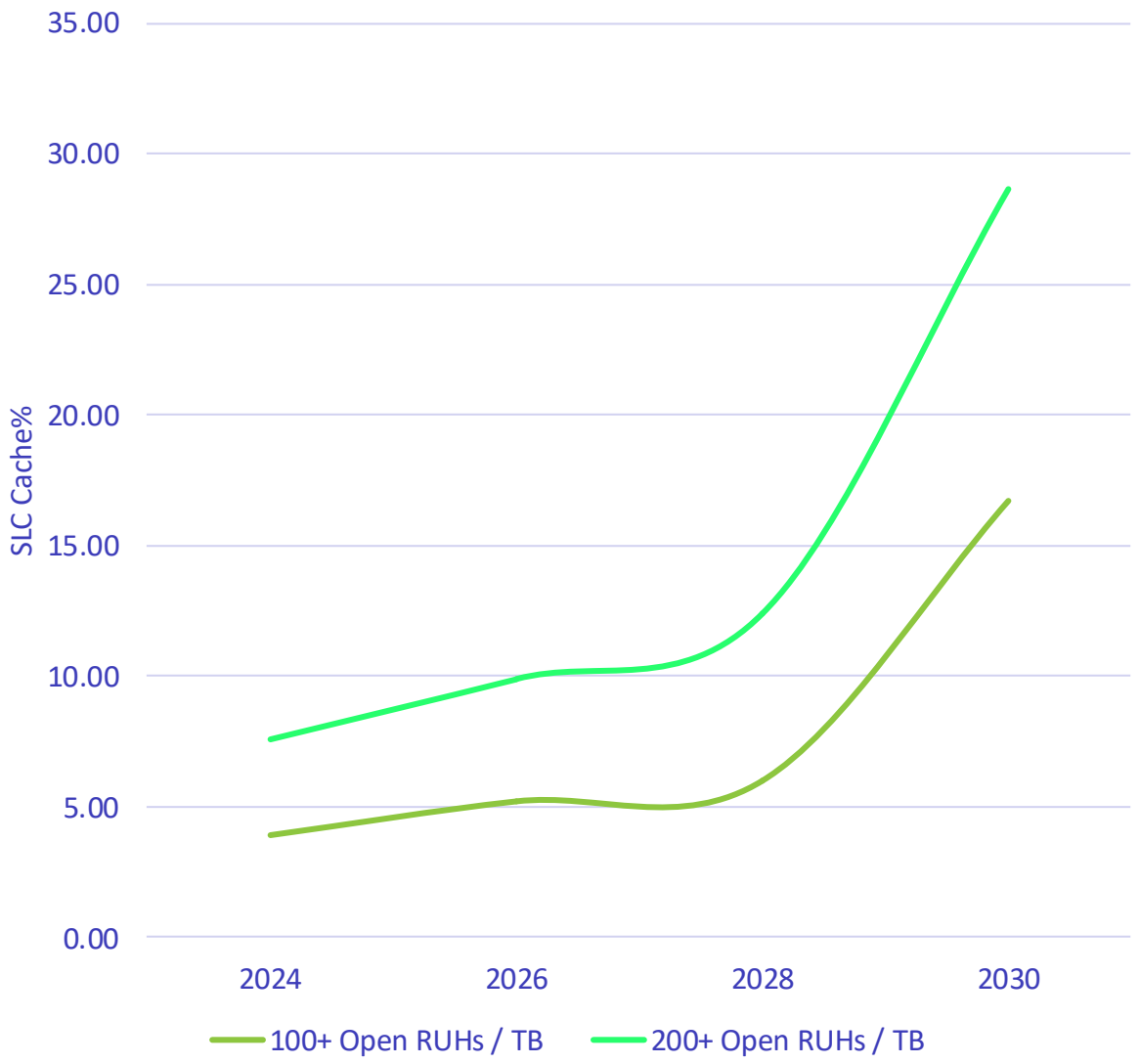
- NAND blocks getting larger:
  - 2D to 3D NAND
  - TLC to QLC
  - Cost efficiency: lower cost per GB
- Sub-blocks:
  - Difficult to use
  - Additional rules required for host to manage
  - Extra complexity



Small RUs Are The Problem

- Customers want to scale performance and tenants with TB:
  - More open blocks as drive scales with small RUs
- Cost of small RU FDP large capacity drives with large number of RUHs will be extremely uncompetitive by mid 2029/2030 timeframe
- Having less RUHs per TB will help but can lessen performance
- Sub-block erase methods can help but require more complex management

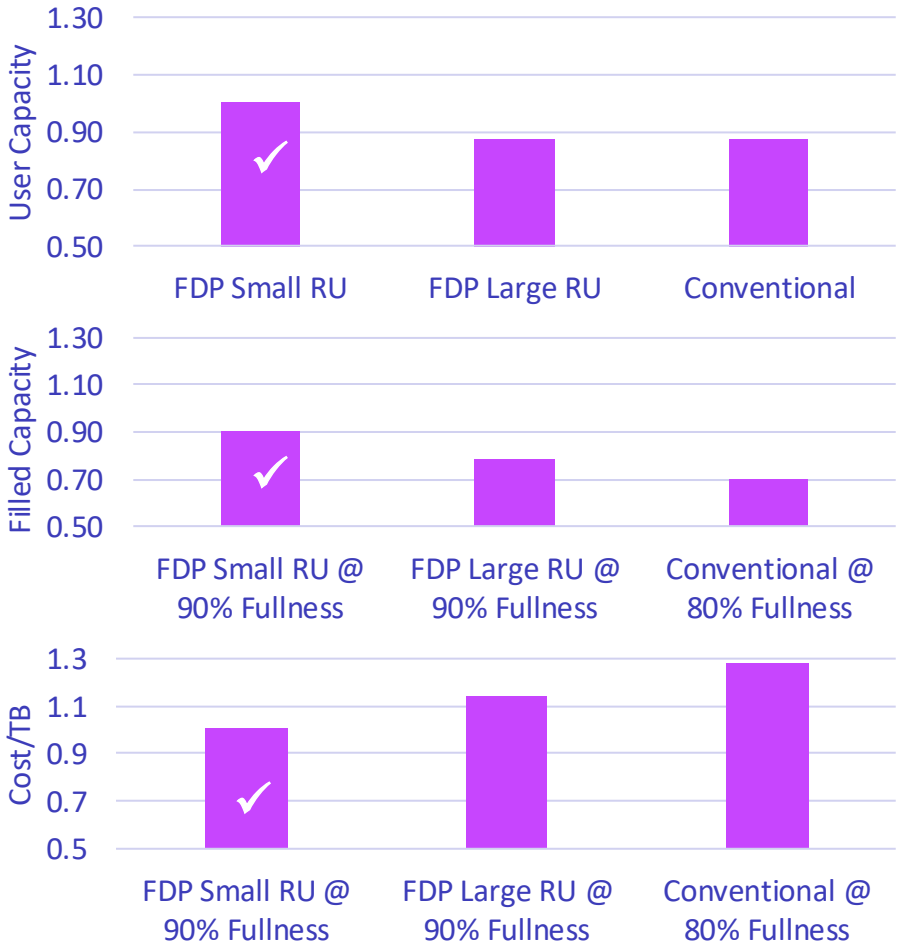
- Avoiding small RUs solves the problem



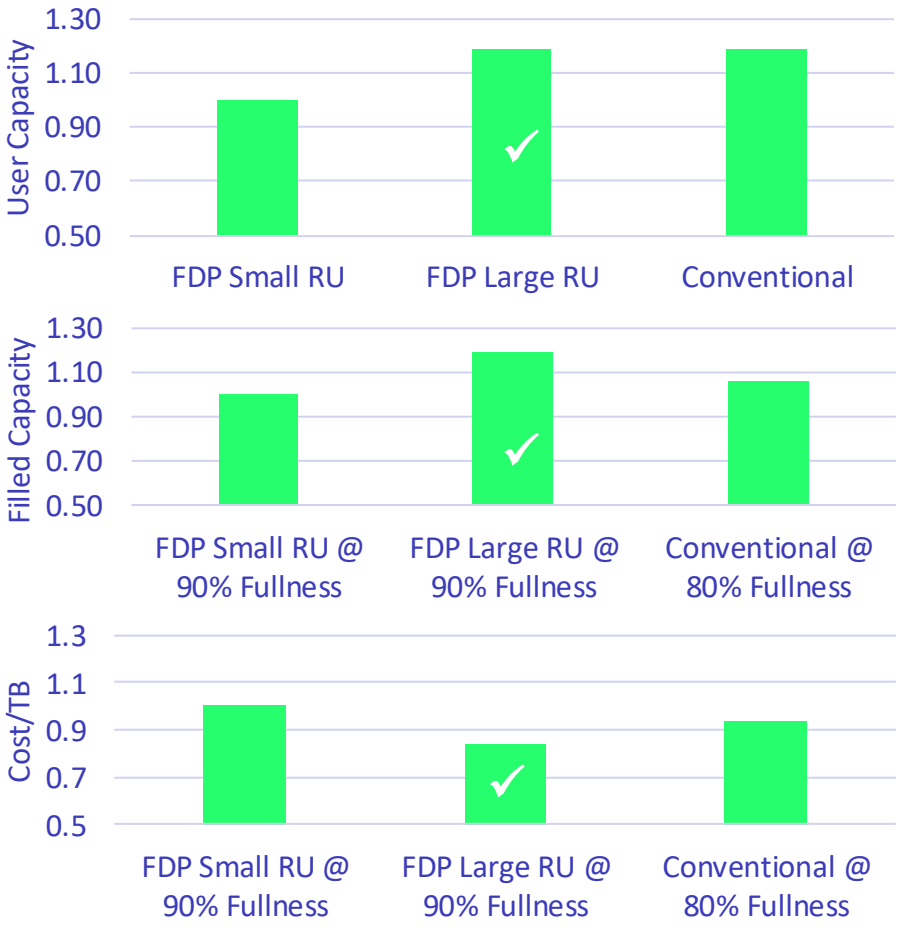
Large RUs Are The Way Forward

FDP Small RU assumes SSD has minimal OP to gain advantage in 2026

2026



2030



Thank You!

